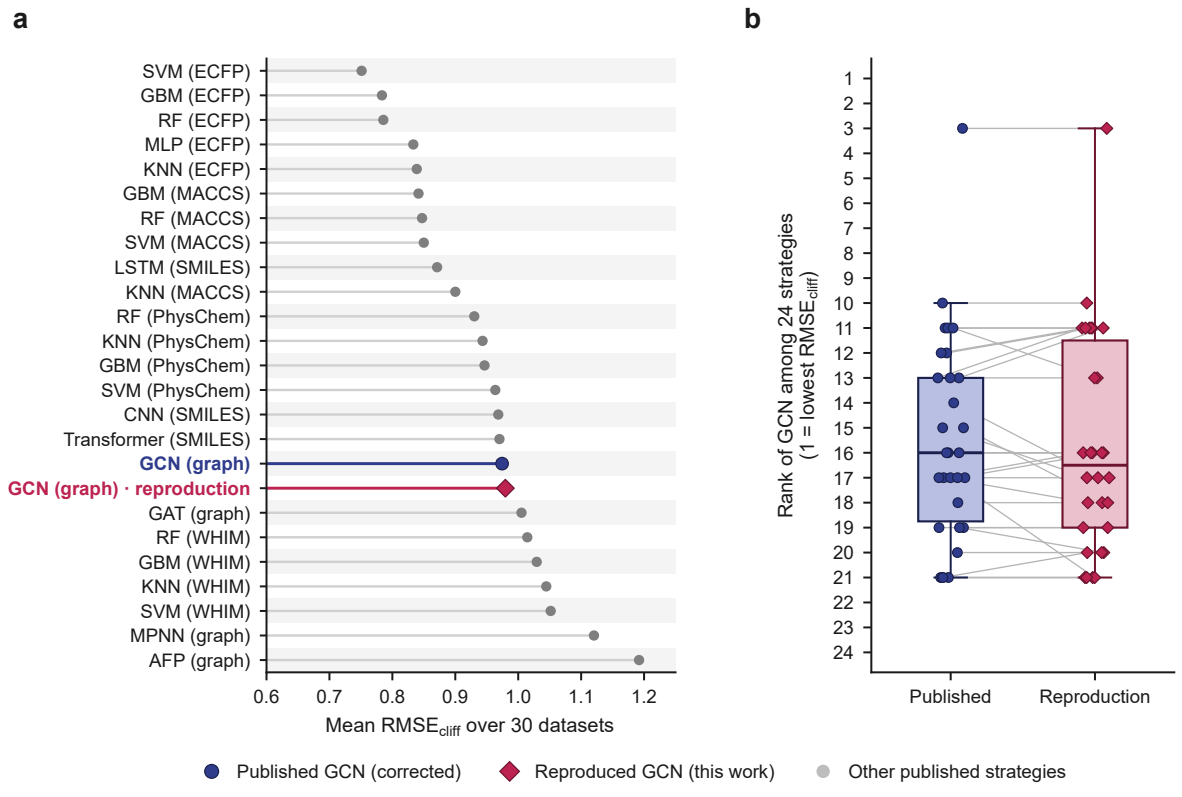

a
SVM (ECFP)
GBM (ECFP)
RF (ECFP)
MLP (ECFP)
KNN (ECFP)
GBM (MACCS)
RF (MACCS)
SVM (MACCS)
LSTM (SMILES)
KNN (MACCS)
RF (PhysChem)
KNN (PhysChem)
GBM (PhysChem)
SVM (PhysChem)
CNN (SMILES)
Transformer (SMILES)
GCN (graph)
GCN (graph) · reproduction
GAT (graph)
RF (WHIM)
GBM (WHIM)
KNN (WHIM)
SVM (WHIM)
MPNN (graph)
AFP (graph)
0.6
0.7
0.8
0.9
1.0
1.1
1.2
Mean RMSE$_{cliff}$ over 30 datasets
b
Rank of GCN among 24 strategies
(1 = lowest RMSE$_{cliff}$)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
Published
Reproduction
Published GCN (corrected)
Reproduced GCN (this work)
Other published strategies